Ian Abuaf Pelo

+39 340-077-7977 | ian.abuaf@alumni.usi.ch | linkedin.com/in/ian-abuaf-pelo | github.com/abuafi | abuafi.github.io

Education

Master of Science in Informatics (MSI) with spec. in Software Development 2022 – Sep. 2024
Università della Svizzera Italiana (GPA: 9.58/10 *summa cum laude,* 123 *ECTS)* *Lugano, Switzerland*

- Obtained a scholarship for best student in faculty (Dec. 2023), with a GPA of 9.68.
- Thesis: **Building an Open-Source Library Network to Identify Dependencies and analysis of license insufficiencies in popular package repositories**, graded 9.5/10. (link)

Bachelor of Science in Informatics (BINF) 2019 – 2022
Università della Svizzera Italiana (GPA: 9.20/10 *summa cum laude,* 153 *ECTS)* *Lugano, Switzerland*

- Thesis: **The medial axis of a convex polygon in expected linear time and generalizations**, graded 10/10.

Liceo Scientifico Scienze Applicate Diploma 2014 - 2019
Istituto Tecnico Industriale Statale "Guglielmo Marconi" (final grade: 70/100*).* *Gorgonzola (MI), Italy*

Experience

PhD student in Software Visualization and Evolution Sept. 2025 – Sept. 2026
REVEAL @ Software Institute – Università della Svizzera Italiana (USI) *Lugano, Switzerland*

- Researched at the REVEAL (Reverse Engineering, Visualization, Evolution Analysis Lab) lab under the supervision of Prof. Dr. Michele Lanza and Dr. Marco Raglianti.
- Created a software visualization layout algorithm based on Poisson Disc Sampling and applied it to various real world software systems.
- Collaborated with the Zephyr Foundation to visualize the evolution of their repository's 10 year history.

Stage as Software Developer Jun. 2025 – Jul. 2025
Banca del Sempione *Lugano, Switzerland*

- Developed a full stack application in ASP.NET Core MVC to collect and store corporate order book data.

Research Assistant for Parsing Open-Source Software Dependencies Jul. 2023 – Aug. 2023
Università della Svizzera Italiana (USI) *Lugano, Switzerland*

- Assisted for the project "Parsing Open-Source Software Dependencies".
- Developed a system to crawl, parse and evaluate open-source software dependencies.
- Wrote over 15K lines of production-ready clean code, performed industry-style code reviews, and worked remotely and independently.

Stage in DAC System for USI Software Atelier 5 Field Project Sep. 2021 - Dec. 2021
DAC System SA *Manno, Switzerland*

- Redesigned frontend GUI for an antenna monitoring system.
- Implemented GUI using the ExtJs framework.

Projects

Various Hackatons and Game Jams | *Python, Godot* Nov. 2021 - Present

- Participated in the Neural Wave (Oct. 2024) and Formula USI (Nov. 2021, 3rd place) Hackatons.
- Participated in various online Game Jams from Jan. 2025 to present, more information on my website abuafi.github.io.

VesselRoute | *Cognos Customvis, Leaflet, TypeScript* Nov. 2024 - Jan. 2025

- Developed a custom visualization for routing sea vessels on the IBM Cognos Analytics platform.
- Published on the public marketplace accelerator.ca.analytics.ibm.com.

Building an Open-Source Library Network to Identify Dependencies (Master Thesis) | *GoLang, Python, Docker, REST, SQLite* Sep. 2023 - Sep. 2024

- Completed the architecture started during the research assistant position.

Technical Skills

Languages: Italian (native speaker), English (C1 level)
Programming: Pharo/Smalltalk, Julia, Ruby, Java, Python, C/C++, C#, JavaScript/TypeScript, GoLang, GLSL, HTML/CSS, SQL, LaTeX
Frameworks: Node.js, ExtJs, Vue.js, Three.js, Spring, ROS2, SDL, LWJGL, LibGDX, Godot
Developer Tools: Git, Linux, Docker, VS Code, Jetbrains IDEs, Blender, Cognos Analytics